EXERCISE 9

Uniqueness and Lipschitz Analysis

Original statement · ProblemsCh2.pdf, p. 2

Existence-uniqueness · Non-Lipschitz points

PROBLEM STATEMENT

Which of the following differential equations have unique solutions (at least on some small time interval) for any non-negative initial condition $x(0) \geq 0$?

(i) $\dot{x} = x(1 - x^2)$,
(ii) $\dot{x} = x^3$,
(iii) $\dot{x} = x^{1/3}$,
(iv) $\dot{x} = x^{1/2}(1 + x)^2$,
(v) $\dot{x} = (1 + x)^{3/2}$.

Source: ProblemsCh2.pdf, Exercise 9 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Each equation is autonomous: $\dot{x} = f(x)$ with initial condition $x(0) = x_0 \geq 0$.

Uniqueness criterion used (Robinson, Theorem 6.2): if f and $\partial f/\partial x$ are continuous in an open neighborhood of (t_0, x_0), then the IVP has a unique solution on some open interval containing t_0. This is a sufficient condition, not a necessary one: a more general sufficient condition is that f be locally Lipschitz in x, $|f(x) - f(y)| \leq L|x - y|$. When the condition fails at a point, uniqueness there must be examined directly.

Part	$f(x)$	$f'(x)$	Where f' is continuous
(i)	$x - x^3$	$1 - 3x^2$	all x
(ii)	x^3	$3x^2$	all x
(iii)	$x^{1/3}$	$\frac{1}{3}x^{-2/3}$	$x \neq 0$
(iv)	$x^{1/2}(1 + x)^2$	$\frac{1}{2}x^{-1/2}(1 + x)^2 + 2x^{1/2}(1 + x)$	$x > 0$
(v)	$(1 + x)^{3/2}$	$\frac{3}{2}(1 + x)^{1/2}$	$x > -1$ (so all $x \geq 0$)

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: for each f, check where f and f' are continuous. On those sets Theorem 6.2 applies. At any point $x_0 \geq 0$ where it does not apply (here only $x_0 = 0$ in (iii) and (iv), where f' is unbounded), decide uniqueness by constructing two solutions (separation of variables) or by showing that they cannot exist.

For a decisive non-uniqueness test at $x_0 = 0$ with $f(0) = 0$ and $f > 0$ for $x > 0$: the constant $x \equiv 0$ is always a solution, and by separation of variables, $t = \int_0^x ds/f(s)$ defines a second solution $x(t) > 0$ for $t > 0$ whenever the integral $\int_0^\varepsilon ds/f(s)$ is finite (the time needed to leave 0 is then finite).

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $\dot{x} = x(1 - x^2)$

$f(x) = x - x^3$ and $f'(x) = 1 - 3x^2$ are polynomials, continuous everywhere. Theorem 6.2 applies at every $x_0 \geq 0$. Answer: unique solution for all $x_0 \geq 0$.

Part (ii): $\dot{x} = x^3$

$f = x^3$, $f' = 3x^2$, continuous everywhere. Answer: unique solution for all $x_0 \geq 0$ (locally in time). Remark: the solution blows up in finite time. Separating, $\int_{x_0}^x s^{-3}\, ds = t$ gives $-\frac{1}{2x^2} + \frac{1}{2x_0^2} = t$, so $x(t) = \left(x_0^{-2} - 2t\right)^{-1/2}$ for $x_0 > 0$, which exists only for $t < T^* = \frac{1}{2x_0^2}$. This is why the problem says 'at least on some small time interval'.

Part (iii): $\dot{x} = x^{1/3}$

$f = x^{1/3}$ is continuous, but $f'(x) = \frac{1}{3}x^{-2/3} \to \infty$ as $x \to 0^+$. For $x_0 > 0$, f and f' are continuous near x_0, so the solution is unique. For $x_0 = 0$ the theorem does not apply, and uniqueness in fact fails: the constant $x \equiv 0$ is a solution, and separating variables with $x \geq 0$,

$$\int_0^x s^{-1/3}\, ds = \left[\tfrac{3}{2}s^{2/3}\right]_0^x = \tfrac{3}{2}x^{2/3} = t \quad \Longrightarrow \quad x(t) = \left(\tfrac{2t}{3}\right)^{3/2}, \quad t \geq 0,$$

is a second solution with $x(0) = 0$. Check: $\dot{x} = \frac{3}{2} \cdot \frac{2}{3}\left(\frac{2t}{3}\right)^{1/2} = \left(\frac{2t}{3}\right)^{1/2}$ and $x^{1/3} = \left(\frac{2t}{3}\right)^{1/2}$. So there are at least two solutions from $x_0 = 0$ (indeed infinitely many, obtained by staying at 0 until any time $c \geq 0$ and then following $\left(\frac{2(t-c)}{3}\right)^{3/2}$). Answer: unique for $x_0 > 0$, not unique for $x_0 = 0$.

Part (iv): $\dot{x} = x^{1/2}(1 + x)^2$

$f(x) = \sqrt{x}\,(1 + x)^2$ is continuous on $x \geq 0$ and $f'(x) = \frac{(1+x)^2}{2\sqrt{x}} + 2\sqrt{x}\,(1 + x) \to \infty$ as $x \to 0^+$. For $x_0 > 0$ Theorem 6.2 applies and the solution is unique. At $x_0 = 0$ the theorem does not apply. We show non-uniqueness: besides $x \equiv 0$, separate variables for $x > 0$ and substitute $s = r^2$, $ds = 2r\, dr$ (so $\sqrt{s} = r$):

$$t = \int_0^x \frac{ds}{\sqrt{s}\,(1+s)^2} = \int_0^{\sqrt{x}} \frac{2\,dr}{(1+r^2)^2} = \left[\frac{r}{1+r^2} + \arctan r\right]_0^{\sqrt{x}} = \frac{\sqrt{x}}{1+x} + \arctan\sqrt{x}.$$

The primitive is checked by differentiation: $\frac{d}{dr}\left[\frac{r}{1+r^2} + \arctan r\right] = \frac{1-r^2}{(1+r^2)^2} + \frac{1}{1+r^2} = \frac{2}{(1+r^2)^2}$. The function $T(x) = \frac{\sqrt{x}}{1+x} + \arctan\sqrt{x}$ is continuous and strictly increasing on $x \geq 0$ (its derivative is $1/(\sqrt{x}(1+x)^2) > 0$), with $T(0) = 0$ and $T(x) \to \pi/2$ as $x \to \infty$. Its inverse $x(t) = T^{-1}(t)$, $0 \leq t < \pi/2$, is a solution with $x(0) = 0$ and $x(t) > 0$ for $t > 0$. So at $x_0 = 0$ there are at least two solutions. Answer: unique for $x_0 > 0$, not unique for $x_0 = 0$.

Part (v): $\dot{x} = (1+x)^{3/2}$

$f = (1+x)^{3/2}$ and $f' = \frac{3}{2}(1+x)^{1/2}$ are continuous for $x > -1$, in particular for every $x_0 \geq 0$. Answer: unique solution for all $x_0 \geq 0$. The solution exists only up to a finite time: separating,

$$\int_{x_0}^{x} (1+s)^{-3/2}\,ds = \left[-2(1+s)^{-1/2}\right]_{x_0}^{x} = -\frac{2}{\sqrt{1+x}} + \frac{2}{\sqrt{1+x_0}} = t,$$

so $x(t) = \left(\frac{1}{\sqrt{1+x_0}} - \frac{t}{2}\right)^{-2} - 1$, which blows up at $T^* = \frac{2}{\sqrt{1+x_0}}$ (for $x_0 = 0$, $T^* = 2$).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Summary of the criterion check and numerical cross-checks: for (iii), with $x(t) = (2t/3)^{3/2}$ the residual $\dot{x} - x^{1/3}$ evaluated at $t = 0.7, 1.3, 5$ is of order 10^{-17} (rounding); for (iv), sympy gives $T'(x)\,f(x) = 1$ identically; for (ii) and (v) the closed forms give residual 0 symbolically.

Non-uniqueness and the Lipschitz condition are consistent: in (iii) and (iv), $\frac{|f(x)-f(0)|}{x} = x^{-2/3}$ and $\sim x^{-1/2}$ respectively, which are unbounded as $x \to 0^+$, so no Lipschitz constant exists near 0. (For (iii) and (iv) the integral $\int_0 ds/f(s)$ converges, which is exactly what allows the solution to leave 0 in finite time.) By contrast, in (i), (ii), (v) f is Lipschitz on every bounded interval, and uniqueness holds.

ANSWER

Part	Unique for every $x_0 \geq 0$?	Comment
(i)	Yes	$f \in C^1$; forward in time the solutions stay bounded (they tend to the equilibria 0 or 1). Backward in time, for $x_0 > 1$ the solution escapes to $+\infty$ in finite negative time, because $\int^\infty dx/(x - x^3)$ converges
(ii)	Yes (locally in time)	blow-up at $T^* = 1/(2x_0^2)$ for $x_0 > 0$
(iii)	No	fails at $x_0 = 0$: $x \equiv 0$ and $x = (2t/3)^{3/2}$
(iv)	No	fails at $x_0 = 0$: $x \equiv 0$ and $x = T^{-1}(t)$

Part	Unique for every $x_0 \geq 0$?	Comment
(v)	Yes (locally in time)	blow-up at $T^* = 2/\sqrt{1+x_0}$